Notes on Spectral Sequences

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February 20th, 2020

Recall that a **subcomplex** D_* of a chain complex C_* is a collection of submodules $D_i \subset C_i$ with intermediary maps ∂^{D_i} defined to be restrictions of the boundary maps "upstairs" in C_* and such that these maps commute with the inclusions $\iota_i : D_i \hookrightarrow C_i$. We have a short exact sequence

$$0 \to D_* \to C_* \to C_*/D_* \to 0$$

This short exact sequence induces a long exact sequence on homology:

$$\cdots \to H_k(D_*) \to H_k(C_*) \to H_k(C_*/D_*) \xrightarrow{\delta} H_{k-1}(D_*) \to H_{k-1}(C_*) \to \cdots$$

We consider for a moment what these homology classes actually are: We have the map

$$\partial : C_k/D_k \to C_{k-1}/D_{k-1}$$

which treats elements of C_k that differ by elements of the submodule D_k as a single class and then maps them into C_{k-1}/D_{k-1} via the induced map of $\partial_{C_k} : C_k \to C_{k-1}$ on the quotient. The kernel of this map consists of those elements of C_k whose boundaries lie in D_{k-1}. The homology group $H_k(C_*/D_*)$ then is this kernel modulo the image of the similarly defined map upstairs:

$$\partial_{C_{k+1}/D_{k+1}} : C_{k+1}/D_{k+1} \to C_k/D_k$$

Since, by construction, any representative $x \in C_k$ of a homology class $\alpha \in H_k(C_k/D_k)$ has $\partial x \in D_{k-1}$, we have a well defined map

$$\delta : H_k(C_*/D_*) \to H_{k-1}(D_{k-1})$$

$$\delta : \alpha \mapsto [\partial x]$$

If it is the case that computing the homology of D_* and C_*/D_* is easier (somehow) than computing that of C_*, then we can compute the homology groups $H_i(C_*)$ by situating them in short exact sequences

$$0 \to coker(\delta|_{H_{k+1}(C_*/D_*)}) \to H_k(C_*) \to ker(\delta|_{H_k(C_*/D_*)}) \to 0$$

A **filtered R-module** is an R-module M with an increasing sequence of submodules $F_pM \subset F_{p+1}M$ whose union is M whose intersection is trivial.

We say a ring R is **graded** if there is a family of subgroups $\{R_n\}_{n\in\mathbb{Z}}$ such that

$$R = \bigoplus_{n\in\mathbb{Z}} R \quad \text{and} \quad R_n \cdot R_m \subset R_{n+m}$$

Given a graded ring R (with subgroups $\{R_n\}_{n\in\mathbb{Z}}$), we say that an R-module M is a **graded R-module** if there is a family of subgroups $\{M_n\}_{n\in\mathbb{Z}}$ whose direct sum is M and satisfies the property:

$$R_n \cdot M_m \subseteq M_{n+m}$$

A filtered chain complex is a chain complex (C_*, ∂) together with a filtration $\{F_pC_i\}$ of each C_i such that $\partial(F_pC_i) \subset F_pC_{i-1}$.

A filtration on a chain complex C_* induces a filtration on its homology which we can define as follows:

$$F_pH_i(C_*) = \{\alpha \in H_i(C_*) \mid \exists x \in F_pC_i : \alpha \in [x]\}$$

We should also observe that any differential preserving the filtration F_pC_* will descend to the quotient, giving an induced map

$$\partial_{quot} : F_pC_i/F_{p-1}C_i \to F_pC_{i-1}/F_{p-1}C_{i-1}$$

$$\partial_{quot}([x]) = [\partial x]$$

It's easy to get lost in all this indexing, it may help to reexamine our work in diagram form. We start with the chain complex:

$$\cdots \longrightarrow C_{n+1} \xrightarrow{\partial} C_n \xrightarrow{\partial} C_{n-1} \longrightarrow \cdots$$

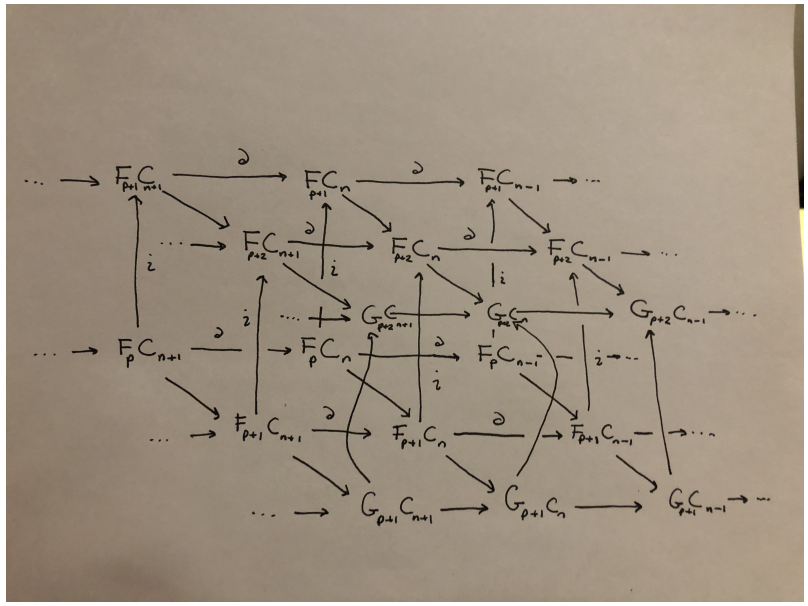
If each C_i is a graded module satisfying $\partial F_pC_i \subseteq F_pC_{i-1}$, then we have a diagram

$$\begin{tikzcd}
 & \vdots & \vdots & \vdots & \\
\cdots \arrow[r] & F_{p+1}C_{n+1} \arrow[r, "\partial"] \arrow[u] & F_{p+1}C_n \arrow[r, "\partial"] \arrow[u] & F_{p+1}C_{n-1} \arrow[r] \arrow[u] & \cdots \\
\cdots \arrow[r] & F_pC_{n+1} \arrow[r, "\partial"] \arrow[u, "\iota"] & F_pC_n \arrow[r, "\partial"] \arrow[u, "\iota"] & F_pC_{n-1} \arrow[r] \arrow[u, "\iota"] & \cdots \\
 & \vdots \arrow[u] & \vdots \arrow[u] & \vdots \arrow[u] &
\end{tikzcd}$$

and now the short exact sequence

$$0 \to F_{p-1}C_* \to F_pC_* \to G_pC_* \to 0$$

gives us the diagram



We observe that since ∂ preserves the filtration of each C_i, its induced map on the quotient, ∂_{quot} will preserve the filtration G_pC_*, meaning that (G_pC_*, ∂) is also a filtered chain complex.

Let (F_pC_*, ∂) be a filtered chain complex. We hereby denote the associate graded sequence by

$$E^0_{p,q} := G_pC_{p+q} = F_pC_{p+q}/F_{p-1}C_{p+q}$$

Just to make sure we're keeping abreast of what's going on here, remember that E^0_{p+q} is really just the portion of F_pC_{p+q} that differs from the submodule $F_{p-1}C_{p+q}$ sitting right below it in the upside-down staircase structure of the filtration. The weird indexing over (p, q) is motivated by a few more things yet to be defined.

Since $\partial : C_* \to C_{*-1}$ respects the filtration structure of our chain complex F_pC_*, we can use it to define an induced differential

$$\partial_0 : E^0_{p,q} \to E^0_{p,q-1}$$

by taking class $[x] \in G_pC_{p+q}$ (where $x \in F_pC_{p+1}$) and sending it to $[\partial x]$. Thus, at each p, we have a complex

$$\cdots \to E^0_{p,q} \xrightarrow{\partial_0} E^0_{p,q-1} \xrightarrow{\partial_0} E^0_{p,q-2} \to \cdots$$

Since we have a differential map, we have homology of the associate graded sequence, denoted

$$E^1_{p,q} := H_{p+q}(G_pC_*) = \frac{ker(\partial_0 : E^0_{p,q} \to E^0_{p,q-1})}{Im(\partial_0 : E^0_{p,q+1} \to E^0_{p,q})} = \frac{ker(\partial_0 : G_pC_{p+q} \to G_pC_{p+q-1})}{Im(\partial_0 : G_pC_{p+q+1} \to G_pC_{p+q})}$$

This is what Hutchings calls a "first order approximation" of the homology of C_*. The idea here is that as we move higher through our filtration F_pC_* we obtain more of C_* and since the homology of the complex C_* determines homology at each stage in the filtration, it determines the homology at the new stuff introduced at a given level of the filtration. This then gives us a sort of graded structure on the homology of C_*.

We observe that a homology class $\alpha \in E^1_{p,q}$ is represented by an element $x \in F_pC_{p+q}$ satisfying $\partial x \in F_{p-1}C_{p+q-1}$, i.e. sent to a portion of C_{p+q-1} made redundant at stage p of the filtration of C_{p+q-1}. We can derive a "second-order approximation" to the homology of C_* by defining

$$\partial_1 : E^1_{p,q} \to E^1_{p-1,q}$$

to be such that it takes the aforementioned homology class $\alpha = [x]$ to the class $[\partial x]$. That is, each element representing a given homology class in $E^1_{p,q}$ represents said class precisely because it

is either killed by ∂ or it is sent to an element which arose at an earlier stage of the filtration. Since ∂ is well-defined, so is ∂_1 and it's clear that since $\partial_1 : [x] \mapsto [\partial x]$, we can see that $\partial_1^2 = 0$. We then define

$$E^2_{p,q} = H_p(E^1_{p,q}) = \frac{ker(\partial_1 : E^1_{p,q} \to E^1_{p-1,q})}{Im(\partial_1 : E^1_{p+1,q} \to E^1_{p,q})}$$

Serre Spectral Sequences

We now look at a particular application of spectral sequences to the computation of topological invariants. We begin with a space X and a filtration on X, that is, a sequence of subspaces

$$\cdots \subset X_p \subset X_{p+1} \subset \cdots \qquad \bigcup_{i \in \mathbb{N}} X_i = X$$

and we'll go ahead and assume that $X_p = \varnothing$ for $p < 0$. Hatcher then makes the diagram pictured below:

$$\begin{tikzcd}
 & \arrow[d] & & \arrow[d] & & \arrow[d] & \\
\arrow[r] & H_{n+1}(X_p) \arrow[r] \arrow[d] & H_{n+1}(X_p, X_{p-1}) \arrow[r] & H_n(X_{p-1}) \arrow[r] \arrow[d] & H_n(X_{p-1}, X_{p-2}) \arrow[r] & H_{n-1}(X_{p-2}) \arrow[r] \arrow[d] & \\
\arrow[r] & H_{n+1}(X_{p+1}) \arrow[r] \arrow[d] & H_{n+1}(X_{p+1}, X_p) \arrow[r] & H_n(X_p) \arrow[r] \arrow[d] & H_n(X_p, X_{p-1}) \arrow[r] & H_{n-1}(X_{p-1}) \arrow[r] \arrow[d] & \\
\arrow[r] & H_{n+1}(X_{p+2}) \arrow[r] \arrow[d] & H_{n+1}(X_{p+2}, X_{p+1}) \arrow[r] & H_n(X_{p+1}) \arrow[r] \arrow[d] & H_n(X_{p+1}, X_p) \arrow[r] & H_{n-1}(X_p) \arrow[r] \arrow[d] & \\
 & & & & & &
\end{tikzcd}$$

The highlighted portion is the long exact sequence of the pair (X_p, X_{p-1}). The diagram effectively encodes information about all the long exact sequences of the sequential pairs in our filtration of X.

There are three conditions on our sequence that will be satisfied by the Serre spectral sequence:

1. All but finitely many of the maps in each $H_k(X_*)$-column are isomorphisms. By exactness this is equivalent to saying that only finitely many terms in each $H_k(X_*, X_{*-1})$ column are nonzero.
2. $H_k(X_{-\infty}) = 0$ for all n
3. $H_h(X_\infty) = 0$ for all n

These conditions state the following: at the top and bottom of column, the groups $H_k(X_p)$ have the common value $H_k(X_{-\infty})$. Since our differentials d_r carry us upward $r-1$ rows, we know that d_r becomes zero for sufficiently large r, which means that the $E^r_{n,p}$ are independent of r for sufficiently large r, we denote these stable values by $E^\infty_{n,p}$.

This structure may also be expressed in the form of a **exact couple**, that is, a diagram of the form

$$\begin{tikzcd}
A \arrow[r, hook, "i"] & A \arrow[dl, "j"] \\
E \arrow[u, "k"] &
\end{tikzcd}$$

which is exact at each stage. Given an exact couple, we should observe that the map $d = k \circ j$ gives us a boundary map since $d^2 = k \circ j \circ k \circ j = k \circ 0 \circ j = 0$ by exactness. With this, we can construct an exact couple created by considering restrictions to the images of your maps, yielding what's known as the **derived couple**

$$\begin{tikzcd} A' \arrow[r, hook, "i'"] & A' \arrow[dl, "j'"] \\ B \arrow[u, "k'"] & \end{tikzcd}$$

where $A' = i(A)$, $i' = i|_{A'}$, $j' = j|_{A'}$, and $E' = ker(d)/Im(d)$.
We now relabel $A^1_{n,p} = H_n(X_p)$ and $E^1_{n,p}(X) = H_n(X_p, X_{p-1})$. The staircase diagram from before is now

$$\begin{tikzcd}
 & \arrow[d] & & \arrow[d] & & \arrow[d] & \\
\arrow[r] & A^1_{n+1,p} \arrow[r] \arrow[d] & E^1_{n+1,p} \arrow[r] & A^1_{n,p-1} \arrow[r] \arrow[d] & E^1_{n,p-1} \arrow[r] & A^1_{n-1,p-2} \arrow[r] \arrow[d] & \\
\arrow[r] & A^1_{n+1,p+1} \arrow[r] \arrow[d] & E^1_{n+1,p+1} \arrow[r] & A^1_{n,p} \arrow[r] \arrow[d] & E^1_{n,p} \arrow[r] & A^1_{n-1,p-1} \arrow[r] \arrow[d] & \\
\arrow[r] & A^1_{n+1,p+2} \arrow[r] \arrow[d] & E^1_{n+1,p+2} \arrow[r] & A^1_{n,p+1} \arrow[r] \arrow[d] & E^1_{n,p+1} \arrow[r] & A^1_{n-1,p} \arrow[r] \arrow[d] & \\
 & & & & & &
\end{tikzcd}$$

If

$$j_1 : A^1_{k,p} \to E^1_{k,p} \qquad k_1 : E^1_{k,p} \to A^1_{k,p-1}$$

we can leverage our consideration of the exact couple to get the differential $d_1 = j_1 k_1$ and then can define the next page in the spectral sequence as follows:

$$A^2_{k,p+1} = i(A^1_{k,p-1}) \qquad E^2_{k,p} = ker(d_1)/Im(d_1)$$

which looks like this:

$$\begin{tikzcd}
 & \arrow[d] & & \arrow[d] & & \arrow[d] & \\
\arrow[r] & A^2_{n+1,p} \arrow[r] \arrow[d] & E^2_{n+1,p} \arrow[r] & A^2_{n,p-1} \arrow[r] \arrow[d] & E^2_{n,p-1} \arrow[r] & A^2_{n-1,p-2} \arrow[r] \arrow[d] & \\
\arrow[r] & A^2_{n+1,p+1} \arrow[r] \arrow[d] & E^2_{n+1,p+1} \arrow[r] & A^2_{n,p} \arrow[r] \arrow[d] & E^2_{n,p} \arrow[r] & A^2_{n-1,p-1} \arrow[r] \arrow[d] & \\
\arrow[r] & A^2_{n+1,p+2} \arrow[r] \arrow[d] & E^2_{n+1,p+2} \arrow[r] & A^2_{n,p+1} \arrow[r] \arrow[d] & E^2_{n,p+1} \arrow[r] & A^2_{n-1,p} \arrow[r] \arrow[d] & \\
 & & & & & &
\end{tikzcd}$$

We can continue this pattern by looking at the rth derived couple. Here we see $E^r_{n,p}$ embedded in the exact sequence

$$E^r_{n+1,p+r-1} \xrightarrow{k_r} A^r_{n,p+r-2} \xrightarrow{i} A^r_{n,p+r-1} \xrightarrow{j_r} E^r_{n,p} \xrightarrow{k_r} A^r_{n-1,p-1} \xrightarrow{i} A^r_{n-1,p} \xrightarrow{j_r} E^r_{n-1,p-r+1}$$

Our conditions imply that eventually, for large enough r, we'll have

$$E^r_{n,p} \cong A^r_{n,p+r-1}/i(A^r_{n,p+r-2})$$

or in other words $i^{r-1}(A^1_{n,p})/i^r(A^1_{n,p-1})$, which is really just a quotient of $A^1_{n,\infty}$. Thus $E^\infty_{n,p}$ is isomorphic F^p_n/F^{p-1}_n.

Proposition. *Given our list of assumptions, the stable group $E^\infty_{n,p}$ is isomorphic to the quotient F^p_n/F^{p-1}_n for the filtration of $A^1_{n,\infty}$ by the subgroups $F^p_n = Im(A^1_{n,p} \to A^1_{n-1,p})$*

In the topological application where we look at the homology groups associated to a filtration of a space X, we have $H_n(X)$ filtered by the groups $F^p_n = Im(H_n(X_p) \to H_n(X))$. The group

$\bigoplus_p F_n^p/F_n^{p-1}$ is called the associated graded group of the filtered group $H_n(X)$. Our proposition then states that the spectral sequence converges to $H_*(X)$, i.e. that

$$\bigoplus_p F_n^p/F_n^{p-1} \cong \bigoplus_p E_{n,p}^\infty$$

We recall that a **fibration** is a map $p : E \to B$ having the homotopy lifting property with respect to any space X, that is, given a homotopy $g_t : X \to B$, if there is a lift $\widetilde{g_0} : X \to E$ of g_0 satisfying $p \circ \widetilde{g_0} = g_0$, then there is a lift $\widetilde{g_t}$ of the entire homotopy.

Supposing we have a fibration $\pi : X \to B$ with B a path-connected CW complex, we can filter X by looking at the preimages of the p-skeleton of B under π.

We recall that X is n-connected if all homotopy groups up through n are trivial. A pair (X, A) is n-connected if the homotopy groups of the pair are trivial up through n.

Since (B, B^p) is p-connected, the homotopy lifting property implies that (X, X^p) is also p-connected, so the inclusion $X_p \hookrightarrow X$ induces an isomorphism on $H_n(-; G)$ if $n < p$. The E^1 term consists of the groups $E_{n,p}^1 = H_n(X_p, X_{p-1}; G)$.

Hatcher explains the change in notation as follows: since the groups $E_{n,p}^1 = H_n(X_p, X_{p-1}; G)$ are non-zero only for $n \geqslant p$, meaning we can perform a reindexing $n = p + q$ and note that our earlier observation translates to

$$E_{p,q}^1 = E_{p+q,p}^1 = H_{p+q}(X_p, X_{p-1}; G)$$

is non-zero when $p \geqslant 0$ and $q \geqslant 0$.

Theorem. *Given $F \to X \to B$ a fibration with B path-connected. If $\pi_1(B)$ acts trivially on $H_*(F; G)$, then there is a spectral sequence $\{E_{p,q}^r, d_r\}$ with*

1. *$d_r : E_{p,q}^r \to E_{p-r,q+r-1}^1$ and $E_{p,q}^{r+1} = ker(d_r)/Im(d_r)$ at $E_{p,q}^r$.*
2. *The stable terms $E_{p,q}^\infty$ are isomorphic to the successive quotients F_n^p/F_n^{p-1} in a filtration*

$$0 \subset F_n^0 \subset \cdots \subset F_n^n$$

   of $H_n(X; G)$.

3. *$E_{p,q}^2 \cong H_p(B; H_q(F; G))$*

Proof. We've already shown that 1 and 2 hold for CW complexes. We now let

$$\Phi_\alpha : D_\alpha^p \to B^p$$

be a characteristic map for e_α^p. Letting $\widetilde{D}_\alpha^p = \Phi_\alpha(X_p)$ be the pullback fibration over D_p^α